EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform

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IN

CSE WITH AI

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1. Introduction

EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform is an AI-powered fleet management system designed to improve the monitoring and operation of electric vehicle fleets. The platform focuses on battery health monitoring, energy consumption, predictive maintenance, charging optimization, and energy-efficient route planning.

The purpose of this test case design is to verify the correctness, functionality, reliability, security, and usability of the proposed EcoFleet AI platform. The testing covers normal, boundary, invalid, and exceptional conditions as required by the assessment.

The test cases are designed to ensure that the system can properly monitor vehicle status, predict battery degradation, recommend charging schedules, optimize routes, generate alerts, and provide useful fleet analytics.

2. Functional Requirements

Req. ID	**Functional Requirement**
FR01	The system shall monitor electric vehicle battery health.
FR02	The system shall monitor vehicle energy consumption.
FR03	The system shall display vehicle status in real time.
FR04	The system shall predict battery degradation using machine learning.
FR05	The system shall predict vehicle maintenance requirements.
FR06	The system shall optimize EV charging schedules.
FR07	The system shall consider charging station availability.
FR08	The system shall recommend energy-efficient routes.
FR09	The system shall use GPS and vehicle performance information for route optimization.

FR10	The system shall provide a real-time fleet monitoring dashboard.
FR11	The system shall generate fleet performance reports.
FR12	The system shall provide alerts for critical battery and vehicle conditions.

3. Non-Functional Requirements

Req. ID	Non-Functional Requirement
NFR01	The dashboard should provide timely updates of vehicle information.
NFR02	The system should be reliable during continuous fleet monitoring.
NFR03	The system should protect vehicle and fleet data from unauthorized access.
NFR04	The system should provide accurate battery and energy information.
NFR05	The system should be easy for fleet managers to use.
NFR06	The system should handle multiple vehicles simultaneously.
NFR07	The system should provide scalable performance as fleet size increases.
NFR08	The system should recover gracefully from sensor, GPS, or network failures.

4. Test Scenarios

The major scenarios to be tested are:

1. Vehicle registration and fleet management
2. Battery health monitoring
3. Energy consumption monitoring
4. Real-time vehicle status monitoring
5. Battery degradation prediction
6. Predictive maintenance
7. Charging schedule optimization
8. Charging station availability
9. Route optimization

10. GPS data handling
11. Traffic-aware route recommendation
12. Dashboard visualization
13. Critical battery alerts
14. Fleet performance reports
15. Invalid or missing sensor data
16. Network/GPS failure handling
17. Multiple vehicle monitoring
18. User authentication and security

5. Detailed Test Cases

For the Actual Result and Pass/Fail Status, leave the fields blank initially and fill them after executing your system. This matches the assignment requirement to document actual results and status.

Test Case 1 – Vehicle Status Monitoring

Field	**Details**
Test Case ID	TC01
Requirement ID	FR03
Test Scenario	Verify real-time vehicle status
Test Type	Normal
Test Steps	1. Login to dashboard. 2. Select a vehicle. 3. View current vehicle status.
Test Data	Vehicle ID: EV001, Status: Active
Expected Result	The dashboard should display the vehicle as Active with updated status information.
Actual Result	___________
Pass/Fail	___________

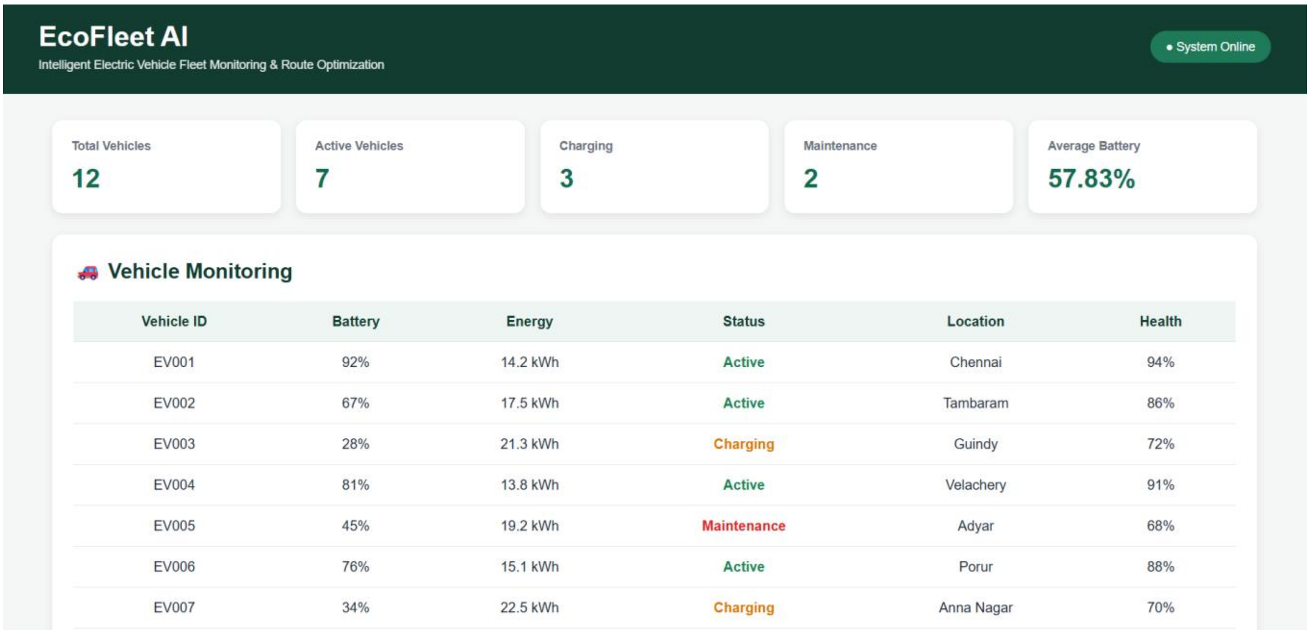


Figure 5.1: EcoFleet AI Fleet Monitoring Dashboard

Test Case 2 – Battery Health Monitoring

Field	**Details**
Test Case ID	TC02
Requirement ID	FR01
Test Scenario	Verify battery health monitoring
Test Type	Normal
Test Steps	1. Select EV001. 2. Open battery information. 3. Check battery percentage and health.
Test Data	Battery level: 80%, Battery health: Good
Expected Result	The system should correctly display battery level and health condition.
Actual Result	____________
Pass/Fail	____________

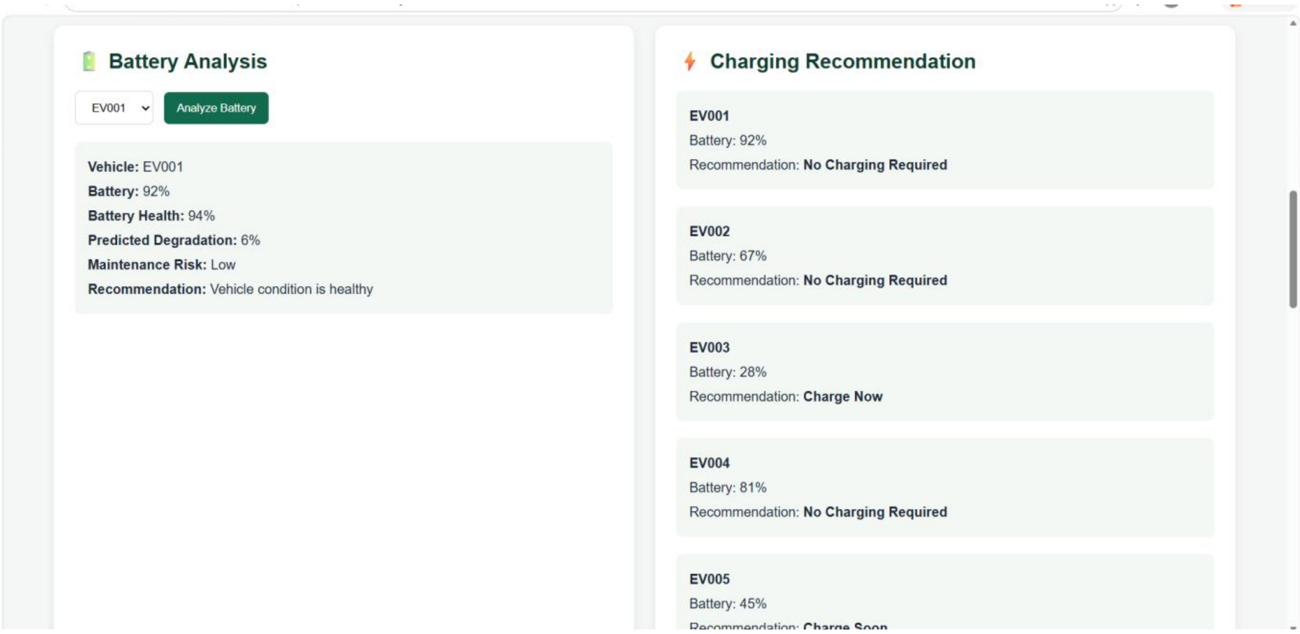


Figure 5.2: Battery Health Monitoring and Alert

Test Case 3 – Low Battery Condition

Field	**Details**
Test Case ID	TC03
Requirement ID	FR12
Test Scenario	Verify low-battery alert
Test Type	Boundary
Test Steps	1. Enter battery level near the configured critical threshold. 2. Monitor dashboard.
Test Data	Battery level: 20%
Expected Result	The system should identify the low battery condition and generate an appropriate alert.
Actual Result	___________
Pass/Fail	___________

Technique: Boundary Value Analysis.

Test Case 4 – Critical Battery Condition

Field	**Details**
Test Case ID	TC04
Requirement ID	FR12
Test Scenario	Verify critical battery alert
Test Type	Exceptional
Test Steps	1. Set vehicle battery to a critically low value. 2. Monitor dashboard.
Test Data	Battery level: 5%
Expected Result	The system should generate a critical battery warning and recommend charging.
Actual Result	___________
Pass/Fail	___________

Test Case 5 – Invalid Battery Value

Field	**Details**
Test Case ID	TC05
Requirement ID	FR01
Test Scenario	Verify handling of invalid battery percentage
Test Type	Invalid
Test Steps	1. Enter battery percentage. 2. Enter an invalid value. 3. Submit the data.
Test Data	Battery level: 120%
Expected Result	The system should reject the invalid value and display a suitable validation message.
Actual Result	___________
Pass/Fail	___________

Technique: Equivalence Partitioning.

Valid_partition:**0–100%**

Invalid partitions: **Below 0% and above 100%**

6. Energy Consumption Test Cases

Test Case 6 – Energy Consumption Monitoring

Field	**Details**
Test Case ID	TC06
Requirement ID	FR02
Test Scenario	Verify energy consumption monitoring
Test Type	Normal
Test Steps	1. Select a vehicle. 2. Start monitoring. 3. View energy consumption.
Test Data	Energy consumption: 18 kWh/100 km
Expected Result	The system should display the vehicle's energy consumption correctly.
Actual Result	___________
Pass/Fail	___________

Test Case 7 – Zero Energy Consumption

Field	**Details**
Test Case ID	TC07
Requirement ID	FR02
Test Scenario	Verify minimum energy consumption value
Test Type	Boundary
Test Steps	1. Provide energy consumption value as zero. 2. Submit the data.
Test Data	Energy consumption: 0 kWh/100 km
Expected Result	The system should accept zero only when the vehicle is stationary or the data is logically valid.
Actual Result	___________
Pass/Fail	___________

7. Battery Degradation Prediction

Test Case 8 – Battery Degradation Prediction

Field	Details
Test Case ID	TC08
Requirement ID	FR04
Test Scenario	Verify battery degradation prediction
Test Type	Normal
Test Steps	1. Provide historical battery data. 2. Run prediction model. 3. View prediction result.
Test Data	Battery history for EV001
Expected Result	The machine learning model should generate a battery degradation prediction.
Actual Result	___________
Pass/Fail	___________

Test Case 9 – Insufficient Battery Data

Field	Details
Test Case ID	TC09
Requirement ID	FR04
Test Scenario	Verify prediction with insufficient data
Test Type	Exceptional
Test Steps	1. Provide incomplete historical battery data. 2. Run prediction.
Test Data	Only 2 historical battery records
Expected Result	The system should not produce an unreliable prediction and should display an appropriate message.
Actual Result	___________
Pass/Fail	___________

8. Predictive Maintenance

Test Case 10 – Maintenance Prediction

Field	**Details**
Test Case ID	TC10
Requirement ID	FR05
Test Scenario	Verify predictive maintenance
Test Type	Normal
Test Steps	1. Upload vehicle performance data. 2. Run maintenance prediction. 3. View result.
Test Data	High battery degradation + increased energy consumption
Expected Result	The system should identify possible maintenance requirements and display a recommendation.
Actual Result	___________
Pass/Fail	___________

9. Charging Optimization

Test Case 11 – Charging Schedule Optimization

Field	**Details**
Test Case ID	TC11
Requirement ID	FR06
Test Scenario	Verify charging schedule optimization
Test Type	Normal
Test Steps	1. Select a vehicle. 2. Enter battery status. 3. Provide charging station availability. 4. Generate charging schedule.
Test Data	Battery: 30%, Station: Available
Expected Result	The system should recommend a suitable charging time and station.

Actual Result	__________
Pass/Fail	__________

Test Case 12 – No Charging Station Available

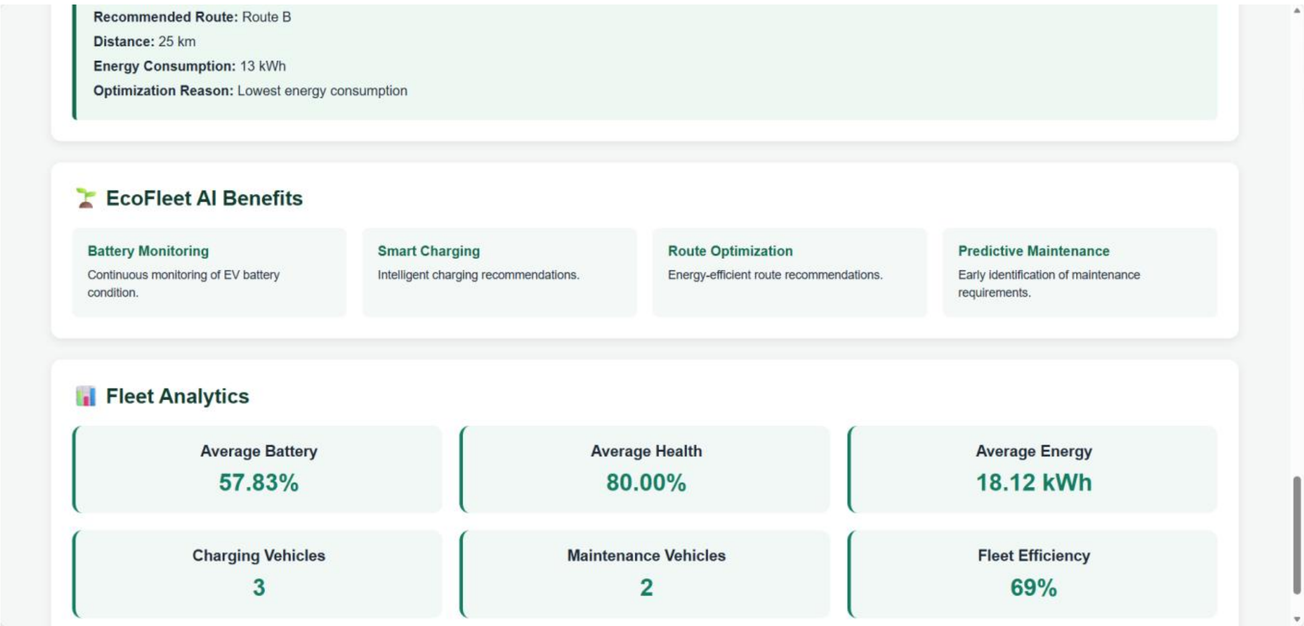
Field	Details
Test Case ID	TC12
Requirement ID	FR07
Test Scenario	Verify charging optimization when stations are unavailable
Test Type	Exceptional
Test Steps	1. Set battery level to low. 2. Mark all nearby charging stations unavailable. 3. Generate charging recommendation.
Test Data	Battery: 15%, Nearby stations: 0
Expected Result	The system should notify the fleet manager and recommend an alternative available charging option if possible.
Actual Result	__________
Pass/Fail	__________

10. Route Optimization

Test Case 13 – Energy-Efficient Route

Field	Details
Test Case ID	TC13
Requirement ID	FR08
Test Scenario	Verify energy-efficient route recommendation
Test Type	Normal
Test Steps	1. Enter starting location. 2. Enter destination. 3. Request route optimization.
Test Data	Source: Chennai, Destination: Bengaluru
Expected	The system should generate a suitable route considering EV energy

Result	efficiency.
Actual Result	___________
Pass/Fail	___________



Test Case 14 – Invalid GPS Location

Field	Details
Test Case ID	TC14
Requirement ID	FR09
Test Scenario	Verify invalid GPS input
Test Type	Invalid
Test Steps	1. Enter invalid GPS coordinates. 2. Request route optimization.
Test Data	Latitude: 200°, Longitude: 250°
Expected Result	The system should reject invalid coordinates and display an appropriate error message.
Actual Result	___________
Pass/Fail	___________

Technique: Equivalence Partitioning.

11. GPS Failure

Test Case 15 – GPS Connection Failure

Field	**Details**
Test Case ID	TC15
Requirement ID	NFR08
Test Scenario	Verify system behavior during GPS failure
Test Type	Exceptional
Test Steps	1. Disconnect GPS source. 2. Continue fleet monitoring.
Test Data	GPS signal unavailable
Expected Result	The system should indicate GPS unavailability and avoid displaying misleading location information.
Actual Result	___________
Pass/Fail	___________

12. Dashboard Testing

Test Case 16 – Dashboard Display

Field	**Details**
Test Case ID	TC16
Requirement ID	FR10
Test Scenario	Verify dashboard information
Test Type	Normal
Test Steps	1. Login to the platform. 2. Open fleet dashboard. 3. Check vehicle information.
Test Data	Fleet containing 5 EVs
Expected Result	Dashboard should display vehicle status, battery level, energy consumption, alerts, and other available fleet information clearly.
Actual Result	___________
Pass/Fail	___________

13. Multiple Vehicle Monitoring

Test Case 17 – Multiple EV Monitoring

Field	**Details**
Test Case ID	TC17
Requirement ID	FR03, NFR06
Test Scenario	Verify monitoring of multiple vehicles
Test Type	Normal
Test Steps	1. Add multiple EVs. 2. Start fleet monitoring. 3. Observe dashboard.
Test Data	10 EVs
Expected Result	The system should display the status of all available vehicles without data mixing or incorrect identification.
Actual Result	___________
Pass/Fail	___________

14. Network Failure

Test Case 18 – Network Disconnection

Field	**Details**
Test Case ID	TC18
Requirement ID	NFR08
Test Scenario	Verify behavior during network failure
Test Type	Exceptional
Test Steps	1. Start fleet monitoring. 2. Disconnect network. 3. Observe system response.
Test Data	Network unavailable
Expected Result	The system should notify the user about connectivity failure and recover when the connection is restored.
Actual Result	___________
Pass/Fail	___________

15. Report Generation

Test Case 19 – Fleet Performance Report

Field	**Details**
Test Case ID	TC19
Requirement ID	FR11
Test Scenario	Verify fleet performance report generation
Test Type	Normal
Test Steps	1. Select reporting period. 2. Select fleet. 3. Generate report.
Test Data	Fleet: EV001–EV010, Period: 30 days
Expected Result	The system should generate a report containing fleet utilization, energy efficiency, and operational performance information.
Actual Result	___________

Pass/Fail	__________

16. Security Testing

Test Case 20 – Unauthorized Dashboard Access

Field	Details
Test Case ID	TC20
Requirement ID	NFR03
Test Scenario	Verify unauthorized access protection
Test Type	Invalid/Security
Test Steps	1. Open dashboard without valid credentials. 2. Attempt to access fleet information.
Test Data	Invalid username/password
Expected Result	The system should deny access and display an appropriate authentication error.
Actual Result	__________
Pass/Fail	__________

17. Test Design Techniques Applied

The assignment specifically recommends techniques such as **Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition Testing**.

17.1 Equivalence Partitioning

Used for dividing input data into valid and invalid groups.

Example: Battery Percentage

Partition	Input	Expected
Invalid	-10%	Reject
Valid	0–100%	Accept
Invalid	120%	Reject

Applied to: **TC05 and TC14**

17.2 Boundary Value Analysis

Used to test values at or near important limits.

Example: Battery Level

Test Value Type

0%	Lower boundary
1%	Near lower boundary
20%	Alert boundary
21%	Above alert boundary
100%	Upper boundary
101%	Invalid boundary

Applied to: **TC03, TC04 and TC05**

17.3 Decision Table Testing

Charging decisions can depend on several conditions.

Battery Level	Station Available	Vehicle Required Soon	Decision
Low	Yes	Yes	Charge immediately
Low	Yes	No	Schedule charging
Low	No	Yes	Find alternative station
High	Yes	No	No immediate charging
High	No	No	Continue operation

This can be used for testing the **intelligent charging module**.

17.4 State Transition Testing

An EV can move through different operational states:

Available → Assigned → Running → Low Battery → Charging → Available

Example:

Current State	Event	Expected Next State
Available	Vehicle assigned	Assigned

Assigned	Trip starts	Running
Running	Battery becomes low	Low Battery
Low Battery	Charging begins	Charging
Charging	Battery sufficiently charged	Available

This verifies that the fleet-management system correctly handles changes in vehicle state.

18. Requirement Traceability Matrix

The rubric gives importance to mapping requirements, scenarios, and test cases for strong traceability and coverage.

Requirement	**Test Cases**	**Coverage**
FR01 – Battery monitoring	TC02, TC05	Covered
FR02 – Energy monitoring	TC06, TC07	Covered
FR03 – Vehicle status	TC01, TC17	Covered
FR04 – Battery prediction	TC08, TC09	Covered
FR05 – Maintenance prediction	TC10	Covered
FR06 – Charging optimization	TC11	Covered
FR07 – Charging station availability	TC12	Covered
FR08 – Route optimization	TC13	Covered
FR09 – GPS/vehicle data	TC14, TC15	Covered
FR10 – Dashboard	TC16	Covered
FR11 – Reports	TC19	Covered
FR12 – Alerts	TC03, TC04	Covered
NFR03 – Security	TC20	Covered
NFR06 – Multiple vehicles	TC17	Covered
NFR08 – Failure recovery	TC15, TC18	Covered

19. Test Case Coverage Summary

Test Category	**Test Cases**
Normal Conditions	TC01, TC02, TC06, TC08, TC10, TC11, TC13, TC16,

	TC17, TC19
Boundary Conditions	TC03, TC07
Invalid Conditions	TC05, TC14, TC20
Exceptional Conditions	TC04, TC09, TC12, TC15, TC18
Security Testing	TC20
Performance/Scalability	TC17
Total Test Cases	**20**

20. Expected Overall Testing Outcome

The testing process is expected to verify that the EcoFleet AI platform correctly monitors EV fleet performance, battery health, energy consumption, charging operations, route efficiency, predictive maintenance, and fleet analytics.

The test cases also verify that the system behaves correctly when it receives invalid inputs, boundary values, missing sensor information, unavailable charging stations, GPS failures, and network failures.

Successful completion of these test cases would provide evidence that the proposed platform is functional, reliable, secure, usable, and capable of supporting intelligent EV fleet management.